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Adiposity and attention deficit hyperactivity disorder comorbidity in Chinese children and adolescents: the role of adiposity duration and screen time

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Abstract

Background This study aimed to investigate the environmental factors associated with attention deficit hyperactivity disorder (ADHD) in Chinese children and adolescents with adiposity (obesity or overweight).

Methods A total of 130 children and adolescents with adiposity and 130 age- and sex-matched controls with normal weight were enrolled in this cross-sectional study. Children with ADHD were identified based on International Classification of Diseases (ICD-10) codes. The severity for psychiatric and behavioral problems, ADHD symptoms, and oppositional defiant disorder was compared between the groups. Multivariable logistic regression analysis was performed to identify factors associated with the comorbidity of ADHD in children and adolescents with adiposity.

Results The detection rate of ADHD was significantly higher in the adiposity group compared to control group (15.4% vs. 6.2%; $p=0.016$). Children and adolescents with adiposity exhibiting higher scores in learning problems, impulsivity/hyperactivity, anxiety, and hyperactivity indices. Exploratory analyses indicated a positive correlation was found between BMI percentile (BMI-P) and inattention symptoms scores in individuals with ADHD ($r=0.512$, $p=0.005$). Logistic regression results showed that factors associated with ADHD in children and adolescents with adiposity included prolonged adiposity duration (> 3 years) (OR = 5.61, 95% CI: 1.79–17.59, $p=0.003$) and excessive screen time (> 3 h/day) (OR = 6.45, 95% CI: 1.59–26.11, $p=0.009$).

Conclusions Prolonged adiposity duration and excessive daily screen time were found to be associated with ADHD among children and adolescents with adiposity.

Keywords Adiposity, Attention deficit hyperactivity disorder, ADHD, Comorbidity, Environmental factors

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Introduction

Over the past decades, the prevalence of childhood adiposity (obesity or overweight) has increased drastically worldwide, raising substantial public health concerns [1]. In China, the rates of overweight and obesity among children aged 6–15 years have reached 19.1% and 12.4%, respectively [2]. Beyond the elevated risk of metabolic and cardiovascular complications, children and adolescents with adiposity display a higher prevalence of psychiatric comorbidities, including attention deficit hyperactivity disorder (ADHD), anxiety, and depression [3].

ADHD is recognized as one of the most prevalent neurodevelopmental disorder in pediatric populations, typified by persistent patterns of inattention, hyperactivity, and impulsivity, which impair academic, social, and family functioning [4]. A meta-analysis, encompassing 67 studies, estimated the prevalence of ADHD to be approximately 6.2% in Chinese children and adolescents, with the inattention presentation being the most common [5]. More recently, a national survey reported that the overall prevalence of ADHD to be 6.4% [6]. Existing evidence suggests a complex interplay between obesity and ADHD [7, 8]. Meta-analytic findings indicate that youth with ADHD have a 30% higher risk of developing obesity later in life compared to their non-ADHD counterparts [8]. Conversely, the prevalence of ADHD is also elevated in children and adolescents with obesity compared to their normal-weight peers [9, 10], reaching as high as 28.6% among severe obese adolescents [9]. Furthermore, early ADHD symptoms may predict subsequent weight gain or obesity, while overweight status may, in turn, exacerbate ADHD symptoms severity [11].

A comprehensive understanding of the co-occurrence mechanisms underlying ADHD and obesity necessitates considering the inherently complex etiology of ADHD itself. ADHD is recognized as a highly heritable disorder, with heritability estimates reaching 76% [12]. However, genetic predisposition is not solely deterministic; its phenotypic expression is significantly modulated by environmental factors, demonstrating the concept of "gene-environment interaction" [7, 13]. Within this integrated framework, acquired factors such as dietary patterns, physical activity, lifestyle habits, and psychosocial stress are not merely direct drivers of weight gain. They may also function as environmental risk factors, interacting with an individual's genetic background, and consequently influencing the onset and progression of ADHD [14].

Currently, research concerning ADHD in Chinese children primarily focuses on the genetic aspects, with comparatively limited exploration of environmental factors. Chinese society exhibits high academic expectations and significant competitive pressure. This context can lead to children experiencing prolonged sedentary study periods, insufficient physical activity, and can even trigger stress-induced eating behaviors [15, 16]. Furthermore, concurrent with rapid economic development, Chinese household dietary patterns are undergoing a "nutrition transition," marked by a sharp increase in the consumption of high-calorie, high-fat, and high-sugar fast food and processed foods [17]. The pervasive social stigma and inadequate public understanding surrounding mental health issues like ADHD can result in delayed diagnosis or misinterpretation of symptoms. Collectively, these culturally specific elements likely shape the distinct manifestations and developmental trajectories of ADHD and obesity in Chinese children and adolescents.

Therefore, this study aims to investigate the occurrence of ADHD among children and adolescents with adiposity in our center. Furthermore, it seeks to analyze the association between obesity status and ADHD symptom presentation, and to identify potential environmental factors linked to ADHD. The findings will provide objective evidence developing more targeted obesity management strategies, contributing to the improved physical and mental well-being of Chinese children and adolescents.

Methods

Study design and participants

This was a retrospective cross-sectional study conducted using data from the Affiliated Hospital of Nantong University between October 2021 and November 2022. The study population was derived from a retrospective screening of electronic medical records (EMR) for children and adolescents diagnosed with overweight or obesity. Inclusion Criteria: (1) Age between 6 and 14 years; (2) Normal intellectual development; (3) Overweight (85th ≤ BMI-*P* < 95th percentile) or obesity (BMI-*P* ≥ 95th percentile) [18]. Exclusion Criteria: (1) Secondary obesity caused by drugs, endocrine or metabolic diseases, and central nervous system lesions; (2) Chronic diseases such as asthma, nephropathy, or thyroid dysfunction; (3) Neurodevelopmental or psychiatric disorders include epilepsy, autism, depression, mania, cerebral palsy, or dementia. The absence of these conditions was determined by a review of the patient's medical history and the lack of a formal ICD-10 diagnosis; (4) Inability to cooperate, poor compliance, or incomplete data.

Ultimately, 130 overweight or obese children (adiposity group) meeting the specified criteria were included. A Control Group ($n = 130$) of age- and sex-matched children with a BMI-P in the normal range (5th ≤ BMI-*P* < 85th percentile) was then selected from the same population covering the same time period.

Diagnosis of ADHD

Children diagnosed with ADHD were identified retrospectively from the standardized medical record system based on International Classification of Diseases (ICD-10) codes. All diagnoses were made by board-certified pediatricians and psychiatrists based on a comprehensive clinical evaluation. Raw data from their assessments were retrieved, including family factor questionnaires, the Swanson, Nolan, and Pelham IV (SNAP-IV) Parent 26-Item Rating Scale, and the Conners Parent Symptom Questionnaire (PSQ).

Assessment tools

Parent Symptom Questionnaire (PSQ)

The PSQ is an internationally recognized and reliable behavioral assessment scale widely used for evaluating ADHD symptoms [19]. This 48-item questionnaire includes five subscales: Conduct Problems, Learning Problems, Psychosomatic Problems, Impulsivity-Hyperactivity, and Anxiety. The psychometric properties of the Chinese version of the PSQ have been validated in prior research, demonstrating test-retest reliability ranging from 0.15 to 0.63, a Cronbach's α of 0.92, and item-total consistency between 0.49–0.85 [20].*Swanson, Nolan, and Pelham IV (SNAP-IV) Parent 26-Item Rating Scale.*

The SNAP-IV scale is structured according to the criteria outlined in the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV). It comprises three subscales: Inattention (items 1–9), Hyperactivity/Impulsivity (items 10–18), and Oppositional Defiant Disorder (ODD; items 19–26). Based on the scores from the ADHD subscales, individuals are categorized into three presentations: Inattentive, Hyperactive/Impulsive, and Combined Presentation. The Chinese version of the SNAP-IV has demonstrated good internal reliability (Cronbach's α = 0.88–0.90) and satisfactory test-retest reliability (intraclass correlation = 0.59–0.72) [21].

Individual and family factors questionnaire

The individual and family factor questionnaires consisted of eleven individual factors related items and four family factors related items. Sleep disorder data were based on guardian-reported physician diagnoses. For initial screen exposure age, daily screen time, and duration of adiposity, specific values were reported by parents or older children and subsequently categorized into two groups. Daily screen time was defined as total daily time spent on electronic devices for both educational and recreational purposes, without distinguishing weekdays from weekends.

Statistical analysis

All statistical analyses were performed using R (version 4.4.2). Continuous variables were tested for normality. Normally distributed data are presented as mean ± SD, independent samples t-tests were used to compare differences between two groups; Non-normally distributed data are presented as median (interquartile range, IQR) and compared using the Mann–Whitney U test. Categorical variables were presented as n(%) and compared using the chi-square test. Pearson correlation coefficients were used to assess correlation between continuous variables. Logistic regression was conducted to identify factors associated with the comorbidity of ADHD in population with adiposity. Variables found to be significantly associated with ADHD in the univariate analysis ($p < 0.05$) were selected for inclusion in the multivariable logistic regression analysis to identify independent factors. A two-sided P-value < 0.05 was considered statistically significant.

Ethical approval and consent

Written informed consent was obtained from guardians of all participants. The study was approved and supervised by the hospital Ethics Committee (approval number: 2022-K017).

Results

In the adiposity group, 67.7% were male and 32.3% were female, with a mean age of 9.72 ± 1.90 years. In the control group, 59.2% were male and 40.8% were female, with a mean age of 9.56 ± 1.97 years. There were no significant differences in sex distribution or age between the two groups ($p = 0.160$) (Table 1).

ADHD was present in 15.4% (20/130) of the adiposity group and 6.2% (8/130) of the control group, with a significant difference between two groups ($p = 0.016$) (Table 1). Among ADHD-diagnosed participants, there was no statistical difference in the distribution of ADHD presentations between those with obesity ($n = 20$) and those with normal weight ($n = 8$) (Table 2).

Table 1 Demographic characteristics and detection rate of ADHD

		Adiposity group ($n = 130$)	Control group ($n = 130$)	t/χ^2	p
Age		9.72 ± 1.90	9.56 ± 1.97	0.643	0.520
Sex	Male	88 (67.7)	77 (59.2)	2.014	0.160
	Female	42 (32.3)	53 (40.8)		
ADHD	Yes	20 (15.4)	8 (6.2)	5.764	0.016*
	No	110 (84.6)	122 (93.8)		

* $p < 0.05$

Table 2 Comparison of ADHD presentations by weight status

	ADHD with Adiposity ($n = 20$)	ADHD with Normal Weight ($n = 8$)	χ^2	p
Inattention	7 (35.0)	4 (50.0)	2.334	0.324
Hyperactivity/impulsivity	2 (10.0)	2 (25.0)		
Combined presentations	11 (55.0)	2 (25.0)		

Severity of psychiatric and behavioral problems, ADHD symptoms and ODD

The severity of psychiatric symptoms was further analyzed among children with ADHD. The results suggested that children with adiposity had higher scores for learning problems, impulsivity/hyperactivity, anxiety, and hyperactivity indices compared to their normal-weight peers ($p < 0.05$) (Table S1). They also showed higher ADHD symptoms scores and ODD scores ($p < 0.05$) (Table S2). Furthermore, an exploratory correlation analysis indicated a positive correlation between BMI-P and inattention scores ($r = 0.512$, $p = 0.005$) (Table S3). Considering the modest sample sizes (adiposity, $n = 20$; normal weight, $n = 8$), the findings from these exploratory analyses should be regarded as preliminary and interpreted with appropriate caution.

Environmental factors of ADHD

Univariate analysis was conducted to examine individual and family environmental factors associated with ADHD within the adiposity group. The results demonstrated that BMI-P, daily screen time, and adiposity duration were significantly higher in the adiposity with ADHD group than in the adiposity without ADHD group ($p < 0.05$). (Table 3).

Logistic regression was performed to identify factors independently associated with ADHD in individuals with adiposity. The analysis revealed that adiposity duration ≥ 3 years (OR = 5.609, 95%CI: 1.79–17.59) and daily screen time ≥ 3 h (OR = 6.451, 95%CI: 1.59–26.11) were both significantly associated with ADHD ($p < 0.05$). BMI-P did not associate with ADHD (OR = 1.01, 95%CI: 0.81–1.25, $p = 0.955$) (Table 4). Hosmer-Lemeshow test indicated an adequate model fit ($\chi^2 = 12.04$, $p = 0.15$), and multicollinearity was not observed among the predictors.

Discussion

This cross-sectional study offers key insights into the comorbidity of adiposity and ADHD in Chinese youth. Our primary finding is that prolonged adiposity duration (≥ 3 years) and excessive daily screen time (≥ 3 h) are significantly associated with the presence of ADHD in children and adolescents with adiposity. These results underscore the importance of considering long-term weight status and lifestyle habits in this population. Furthermore, the overall detection rate of ADHD was higher in the adiposity group compared to the control group (15.4% vs. 6.2%). The result aligns with previous study and meta-analysis [9, 10], highlighting the strong clinical association between these two conditions.

Our findings reveal that individuals with adiposity and ADHD exhibited more pronounced learning difficulties, impulsivity, hyperactivity, and anxiety compared to their normal-weight peers with ADHD ($p < 0.05$). Additionally, the ADHD and ODD symptoms appeared more severe in participants with adiposity ($p < 0.05$). This observation is consistent with the work of Sönmez et al. [22] who reported an association between obesity in children and increased behavioral problems and ADHD symptoms. However, our results should be interpreted with caution due to the small sample size. Nevertheless, these results suggest a potential relationship between adiposity and the severity of psychiatric and behavioral health. Such a relationship could be partially mediated by neurobiological factors. Existing study [23] suggests that excessive weight may be associated with gray matter loss in key brain areas, which could be related to atypical psychosocial development, though the precise mechanisms and directionality require further dedicated investigation.

Our exploratory analysis also suggested a positive correlation between BMI-P and inattention scores in children with ADHD. This preliminary observation contrasts with Hughes et al. [24], who reported no significant association between BMI and ADHD symptoms in a large sample of eight-year-olds. One proposed explanation from that study was the greater impact of family factors on mental health at that age [24]. It has been hypothesized that Inattention is linked to impairments in impulse control and planning [25], which in turn may be associated with dysregulated eating and other lifestyle habits (regular physical activity, healthy dietary choices) [26]. Substantial further research with larger cohorts is essential to validate this potential correlation and determine its clinical significance.

A significant finding from our study was that among children and adolescents with adiposity, those who reported three or more hours of daily screen-related activities had 6.45 times the odds of having an ADHD diagnosis (OR = 6.45, 95% CI: 1.59–26.11, $p = 0.009$) compared to their peers with lower screen exposure. This result is consistent with previous studies. Lee et al. [27] observed a positive relationship between screen time, BMI, and waist circumference. Lingineni et al. [28] also found that children with ADHD symptoms spent significantly more time on screens than their peers. Furthermore, a meta-analysis [29] showed an association between media use and ADHD symptoms.

The potential mechanisms underlying this observed association are multifaceted. Excessive screen time is often associated with reduced physical activity. In China, where intense academic competition and the pervasive availability of high-quality online educational content are prominent, children's "educational" screen time has surged. Concurrently, the widespread appeal of social media and Tik-Tok has drastically reduced their engagement in sports and outdoor activities. This reduction in physical activity is particularly concerning, as evidence [30, 31] indicates that insufficient physical activity is

Table 3 Characteristics of children with adiposity, stratified by ADHD

Factors	Adiposity with ADHD ($n=20$)	Adiposity without ADHD ($n=110$)	t/χ^2	p
Age(y)	10 (9.0–11.0)	10 (8.0–11.0)	0.464	0.644
Sex			0.064	0.810
Male	14 (70.0)	74 (67.3)		
Female	6 (30.0)	36 (32.7)		
BMI-P (%)	98.6 (98.3–99.1)	97.4 (95.6–99.0)	1.982	0.049*
Complications during pregnancy			2.423	0.119
Yes	8 (40.0)	23 (20.9)		
No	12 (60.0)	87 (79.1)		
Mode of delivery			0.528	0.469
Vaginal	3 (15.0)	28 (25.5)		
Cesarean	17 (85.5)	82 (74.5)		
Term infant			1.084	0.298
Yes	20 (100.0)	99 (90.0)		
No	0 (0.0)	11 (10.0)		
Normal Birth weight			1.186	0.278
Yes	15 (75.0)	96 (87.3)		
No	5 (25.0)	14 (12.7)		
Feeding method			2.254	0.279
Breast	13 (65.0)	56 (51.0)		
Artificial	2 (10.0)	27 (24.5)		
Mixed	5 (25.5)	27 (24.5)		
Initial age of screens exposure			1.226	0.269
< 3years	6 (30.0)	21 (19.1)		
≥ 3years	14 (70.0)	89 (80.9)		
Daily screen time (h)			8.247	0.004*
< 3 h	3 (15.0)	55 (50.0)		
≥ 3 h	17 (85.0)	55 (50.0)		
Sleep disorder			< 0.001	1.000
Yes	12 (60.0)	64 (58.2)		
No	8 (40.0)	46 (41.8)		
Adiposity duration (years)			9.556	0.002*
< 3years	6 (30.0)	74 (67.3)		
≥ 3years	14 (70.0)	36 (32.7)		
Family environment			0.001	0.972
Harmonious	19 (95.0)	101 (91.8)		
Conflict	1 (5.0)	9 (8.2)		
Family income (CNY/month)			0.435	0.915
< 5000	0 (0)	9 (8.2)		
5000–10,000	9 (45.0)	36 (32.7)		
10,000–20,000	7 (35.0)	45 (40.9)		
> 20,000	4 (20.0)	20 (18.2)		
Parents' Education			7.823	0.053
Junior high school or below	2 (10.0)	29 (26.4)		
Senior high school	5 (25.0)	21 (19.1)		
Associate degree	7 (35.0)	50 (45.4)		
Bachelor's degree or above	6 (30.0)	10 (9.1)		

* $p < 0.05$; BMI-P- BMI percentile

bidirectionally association with ADHD symptoms in adolescents, and that regular aerobic exercise may help alleviate these symptoms through neurobiological mechanisms. Secondly, beyond mere duration, some studies suggest that exposure to age-inappropriate media content may be associated with poorer psychosocial outcomes [32, 33]. Repeated exposure to highly stimulating or rapidly paced media content may be linked to change in children's arousal thresholds, which could be related to aggressive behaviors, inattention, and hyperactivity [33].

Table 4 Factors associated with ADHD in children and adolescents with adiposity

	β	OR (95%CI)	*p*
Adiposity duration (years)	1.724	5.61(1.79, 17.59)	0.003*
Daily screen time (h)	1.864	6.45(1.59, 26.11)	0.009*
BMI-P (%)	0.006	1.01(0.81, 1.25)	0.955

* $p < 0.05$

Furthermore, the relationship between screen exposure and ADHD symptoms may be bidirectional. Screen time could also be a correlation or even an outcome of pre-existing attentional challenges [32].

Our study observed an association between sustained adiposity and the presence of ADHD symptoms, particularly when adiposity persisted for three years or more. While multiple studies have explored the complex and interactive association between adiposity and ADHD, limited research has examined the influence of adiposity duration. For instance, Khalife et al.'s longitudinal cohort study [31] discussed the trajectory of weight changes across ages. Cortese et al.'s meta-analysis [8] addressed obesity persistence at various developmental stages. However, neither explicitly investigated the relationship between the adiposity duration and ADHD. We propose that persistent adiposity during critical developmental periods, especially exceeding three years, may be linked to more adverse behavioral, neurodevelopmental, or lifestyle outcomes associated with ADHD. Such circumstances could be associated with challenges in children's neurocognitive and social development. It remains unclear whether adiposity duration is independently associated with ADHD development or acts through mediating factors. Further longitudinal studies are required to clarify these relationships.

The associations observed in our study suggest that for children with adiposity, addressing lifestyle factors like screen time and physical activity may be an important component of a comprehensive management plan. Pediatricians, during routine physical examinations and weight management consultations, could consider preliminary inquiry or observation for ADHD-related presentations, especially in children with prolonged adiposity. This initial attention can facilitate earlier identification of children who might require further evaluation. Such early detection provides an opportunity for potential diagnosis and intervention.

This study examines the factors associated with the co-occurrence of adiposity and ADHD in Chinese children and adolescents, an understudied population. However, there are limitations to consider. As a single-center study with limited participant, the generalizability of our findings is restricted, requiring cautious interpretation. Recalling bias and uncontrolled residual confounding factors (such as genetic susceptibility) may affect the study results. In addition, the study was cross-sectional in nature, and odd radio reflects the relative prevalence of the disease in the two groups, not a strict representation of risk incidence or causal effect. Longitudinal studies integrating neurocognitive and lifestyle measures are therefore needed to clarify the bidirectional relationship between ADHD and adiposity and to establish causality.

Conclusion

This cross-sectional study reveals that the detection rate of ADHD among obese children and adolescents in Nantong, China, is significantly higher than that in their normal-weight peers. Within the obese population, a longer duration of adiposity (≥ 3 years) and excessive daily screen time (≥ 3 h) are significantly associated with the presence of ADHD.

Abbreviations

ADHD	Attention Deficit Hyperactivity Disorder
BMI-P	BMI Percentile
DSM-5	Statistical Manual of Mental Disorders, Fifth Edition
IQR	Interquartile Range
ODD	Oppositional Defiant Disorder
PSQ	Conners Parent Symptom Questionnaire
SNAP-IV	Swanson, Nolan, and Pelham IV Parent 26-Item Rating Scale

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

Dandan Bai and Qi Gu were the principal investigators and primary initiators of this project. Youjia Wu and Jin Cai were responsible for the study design, project planning, progress monitoring, and management. Feifei Shen participated in the conceptualization and design of the study. Guihai Suo and Yuqin Zheng undertook data collection and management. Haiying Li performed the statistical analyses. Dandan Bai and Qi Gu prepared the initial manuscript draft. Feifei Shen, Guihai Suo, Yuqin Zheng, Haiying Li, and Hong Wang critically reviewed the manuscript and provided substantive revisions. All authors have read and approved the final version of the manuscript.

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Data availability

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This observational study was conducted in accordance with the ethical principles of the Declaration of Helsinki. Written informed consent was

voluntarily obtained from all participants' legal guardians after a full explanation of the study procedures. The research protocol was reviewed and approved by the Ethics Committee of Affiliated Hospital of Nantong University (Approval No. 2022-K017).

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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